

# **Social Media Addiction, Self-Regulated Learning, and Academic Procrastination**

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## CHAPTER 1: INTRODUCTION

### 1.1 Background and Motivation

As students we are equipped with responsibilities these responsibilities might take the form of weekly assignments, quizzes, or even preparing for a presentation. It has been found that students nowadays, on average struggle to maintain their schedule and are unable to finish work on time. There are numerous of factors involved that could lead to academic procrastination which positively give rise to procrastination. One of them is social media addiction, with the invention of smartphones and social media platforms. Students spent a significant amount of time on such apps, which negatively affects their overall academic performance. This not only makes them lose valuable time but is also affecting their social relationships, furthermore it weakens concentration, lowers the motivation to learn among students. Research has found that the average teenager that spends about 3 to 6 hours on their phones each day are susceptible to develop mental health problems that include anxiety, depression, and insomnia(Nurmaidah et al., 2025a). It is advised for students to lower their social media usage to 2 hours per day. Although it has been shown that social media addiction directly influences academic procrastination, it has also been seen that not all students who use social media procrastinate. Indicating that there exists another variable that there exists another variable that affects the relationship between social media addiction and academic procrastination. Self-regulated learning serves as a mediating variable in this relationship, it can be called as the commitment one might possess to learn something on their own. Therefore, disciplined students who effectively take out a fixed amount time to study each day, are not affected by procrastination even if they are active on social media. In contrary, to the students that do not spend their time learning anything, tend to get distracted more easily, and thus have a lower self-regulating learning variable. Studies have also shown that self-regulated learning can moderate the effect of social media on procrastination, thus acting as a shield.

The previous studies were purely focused on studying these variables independently, however there is a need to understand how social media addiction can weaken self-regulated learning ability among the students.

## **1.2 Problem Statement**

This study was aimed at studying the extent to which social media addiction can have on academic procrastination. While students are expected to engage in activities that can increase their self-regulating learning ability, in order to manage their academic responsibilities. The accessibility of social media makes it a challenge for the students to remain focused without getting distracted that easily undermines their ability to monitor their study habits.

Although the negative effects of procrastination on academic performance and mental health well-being are known. Therefore, a gap remains to figure out the extent to which social media addiction directly causes procrastination and to also find the extent to which social media addiction can weaken the self-regulating learning ability among students.

## **1.3 Theoretical Framework**

### **Social Media Addiction**

Students can develop social media addiction which will affect their ability to fulfil their daily responsibilities and their ability to learn. They lose their ability to concentrate, (pay attention), and fail to complete important tasks. Students spend a large amount of time on social media platforms such as Instagram, TikTok, Facebook and WhatsApp, YouTube.

causes:

- Lack of Concentrate
- Psychological Effects:
- lower Academic Performance
- low quality of work

### **Self-Regulated Learning**

SRL is the process by which learners take ownership of their educational activities. They plan, monitor, and control their own learning.

Characteristics of SRL Students

- Planning
- Monitoring
- Adjusting/Regulating

### **Academic Procrastination**

Academic procrastination is when a student does not complete all his/her required schoolwork in a timely manner (such as procrastination), despite having a good understanding of the potential negative consequences associated with that. Due to being a result of academic procrastination, students frequently encounter higher levels of stress due to having insufficiently much time to complete assignments by means of completing their assignments at the very last minute; therefore, academic performance levels are significantly lower for these students.

Main characteristics:

- Delayed Academic Tasks
- Stress and Anxiety
- Lower Academic Performance

Academic procrastination caused by:

- Distractions
- Poor Self-Regulated Learning Behaviors
- Low Motivation/Fear of Failure

#### **1.3.1 Social Media Addiction → Academic Procrastination**

Students tend to procrastinate when they are using social media because their time on social media is immediate entertainment, and engagement, while studying may not seem as interesting as social media, nor does it create the feeling of urgency.

**Example:**



A student plans to study for two hours, however most of the time is spent on social media. The time spent on social media has caused them to procrastinate on completing their assignments. Students will have to rush to meet the deadline of their work. As a result, their work will be of a lower quality.

### **1.3.2 Social Media Addiction → Self-Regulated Learning**

A large amount of time spent on social media can affect the student's ability to exercise self-regulation learning. Students that spend a lot of time and attention to activities occurring on their social network. That is way Students may have very little time left to plan, organize and monitor the completion of their study-related tasks. Therefore, they are unable to effectively manage and control their own learning activities.

### **1.3.3 Self-Regulated Learning → Academic Procrastination**

Self-regulated learning (SRL) can help improve the way a student performs academically by enabling them to develop both self-discipline and motivation. Setting both short-term and long-term, achievable, realistic, and attainable goals will help the student create self-discipline by helping to schedule when they study and help evaluate their performance. The combination of these three aspects will allow the student to have a high degree of organization, be focused, be committed to their work, and will help them overcome the tendency to procrastinate.

#### **Example:**

An SRD student will develop a weekly study schedule and then keep track of how they prepared for their studies each day as to how effective the study methods were as to whether they found it easy or difficult to learn the content of the material.

### **1.3.4. Mediating Variable Self-Regulated Learning**

The link between social media addiction and procrastination is mediated by one variable: self-regulated learning (SRL). To clarify, SRL explains how and why someone would procrastinate because of being addicted to social media; in this case, SRL will provide a connection for us.

#### **Example:**

if a student spends several hours on TikTok each day. It becomes difficult for students to create an effective plan to study and monitor progress. Therefore, when the student has weak self-regulation (SRL) skills ultimately it creates strong relationship between social media addiction and academic procrastination.

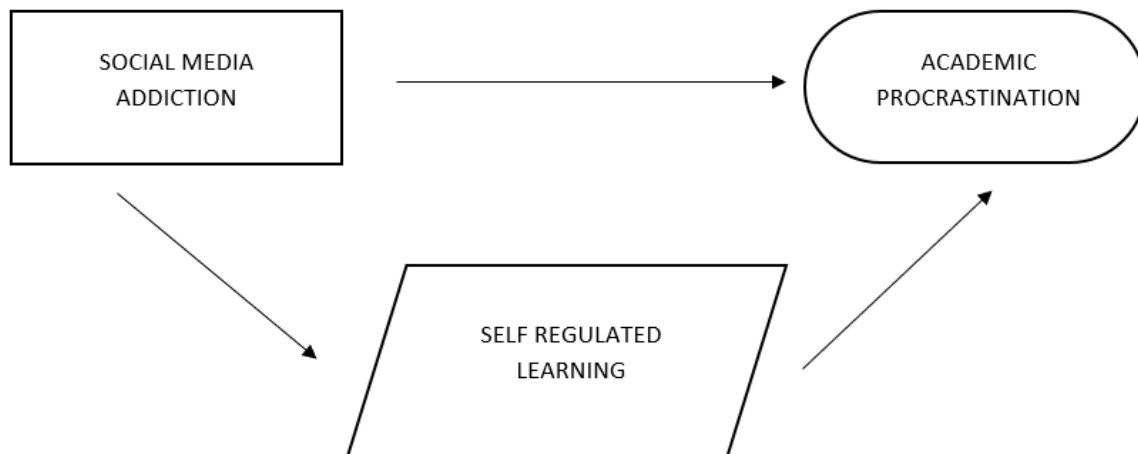
## 1.4 Research objectives

There are different types of objectives in research; these can include high-level objectives and low-level individual objectives. High-level objectives provide the key measure of overall success. While low-level individual objectives should be measurable and achievable steps towards achieving the goals of research.

### Explanation:

It is a study of how social media addiction (X) relates to Academic procrastination (Y) and how self-regulated learning (Z) mediates the relationship between (X) and (Y).

### Partial Mediation:



- To identify how social media addiction (X) directly affects Academic procrastination when studying at university (Y)
- To identify how social media addiction (X) directly affects self-regulated learning (Z)

- To identify how self-regulated learning (Z) directly affects Academic procrastination when studying at university (Y)
- To identify whether self-regulated learning (Z) mediates the relationship between social media addiction (X) and Academic procrastination (Y)

## 1.5 Research Questions

**Q.** What is the effect of Social Media Addiction on Academic Procrastination?

**Q.** What is the effect of Social Media Addiction on Self-regulated Learning?

**Q.** What is the effect of Self-regulated Learning on Academic Procrastination?

**Q.** What is the mediating role of Self-regulated Learning on the relationship of Social Media Addiction and Academic Procrastination?

(Nurmaidah et al., 2025a)

## Hypotheses

**H<sub>0</sub>**= Social Media Addiction has no positive and significant effect on Academic Procrastination.

**H<sub>1</sub>**= Social Media Addiction has positive and significant effect on Academic Procrastination.

**H<sub>0</sub>**= Social Media Addiction has no negative and significant effect on Self-regulated Learning.

**H<sub>2</sub>**= Social Media Addiction has negative and significant effect on Self-regulated Learning.

**H<sub>0</sub>**=Self-regulated Learning has no negative and significant effect on Academic Procrastination.

**H<sub>3</sub>**=Self-regulated Learning has negative and significant effect on Academic Procrastination.

**H<sub>0</sub>**=There is no indirect influence of Social Media Addiction on Academic Procrastination through Self-regulated Learning.

**H<sub>4</sub>**= There is indirect influence of Social Media Addiction on Academic Procrastination through Self-regulated Learning.

(Nurmaidah et al., 2025a)

## 1.6 Contribution of the Study

This study is very important as it shows how self-regulated learning affects the connection between social media addiction and low academic performance. These days many students waste their time on social media due to which they are unable to focus on studies which leads to bad grades. It shows that when a student has good self-control, they are usually better at studies. The psychological theories behavioral theory and learning theory are also applied to the lives of students in this study to help better understand the behavior of students in the online world as the digital technology now plays a big role in a student's life.

However, it is very important to note that social media is not the only reason why students delay studies knowing the negative consequences of it. These studies help researchers understand that sometimes the main issue that leads to academic procrastination is that many modern students don't have strong self-regulated learning skills. And if they did have these skills, they would be able to use social media effectively while being good academically.

In addition to that, this study is very useful for future researchers as they can perform researches on topics linked with student performances, online technologies etc. and use the results of this study.

## 1.7 Significance of the Study

This Study is significant as it can be used by students, teachers, schools/universities as well as other researchers.

**For students:** This is a great resource for students. Many students today are unable to focus on their studies and waste most of their time scrolling on social media trying instead of using their time productively.

By understanding self-regulated learning, students can learn how to:

- control their time
- stay focused on their goals
- avoid unnecessary distractions

- complete work on time

**For teachers:** This study is useful for teachers too. Many teachers find their students distracted, lazy and non-serious towards studies. This research helps teachers understand that the problem is not only social media, but also the lack of self-control in learning.

The results of this study can help teachers create better teaching strategies. For example:

- giving students time-management exercises
- teaching study skills
- encouraging goal setting
- helping students reduce distractions

**For institutions:** This study is very important of schools, universities, tuition centers etc. as social media is growing among students rapidly and educational institutions need strict rules and programs to help guide students in using social media in a responsible way and avoid academic procrastination.

The study can help institutions make better policies related to:

- digital literacy
- responsible use of social media
- online behavior
- time management programs

**For future researchers:** This study can be used by future researchers if they want to research fields such as student lives, education, and digital technology.

They can use this study to explore more topics such as:

- impact of mobile phones on learning
- online learning behavior

- gaming addiction and academic performance
- mental health and procrastination

## **1.8 Conclusion**

In this chapter, we introduce our research by highlighting a real problem that exists today: too much social media causing students to procrastinate. Our early research has shown that 72% of students exhibit high procrastination levels, which we explore in the above chapter. The above research helped us understand the importance of researching the relationship between social media addiction and procrastination.

We developed a theory to illustrate the way three components interact with one another. Addiction to social media causes students to procrastinate, and self-regulated learning helps them regulate their behaviors and utilize time effectively. As our problem statement identified, there is a knowledge gap regarding self-regulated learning's role as a mediating variable between the two constructs for vocational students who learn differently from general high school students.

From this, we were able to come up with four distinct research questions and their associated hypotheses to shape our study. We were able to establish that social media addiction leads to academic procrastination, social media addiction reduces self-regulated learning, and that self-regulated learning reduces procrastination and acts as a mediator.

We also discussed the importance of this research in real life. Students can improve self-control, teachers can improve methods, schools can improve policies, and other researchers can study other interesting topics, such as gaming addiction or online learning behaviors. Therefore, we can see the importance of this research, aside from fulfilling academic requirements.

This chapter gives a clear picture of our entire research project. It identifies the problem, states what we want to prove, and explains why we are doing this research. With this good beginning, we can proceed to the literature review and methodology in the coming chapters.

## CHAPTER 2: CRITICAL LITERATURE REVIEW

### 2.1 Introduction:

The rise of social media platforms with rapid advancements in digital technology, has greatly impacted the lifestyle and academics of most young individuals. As smartphones have become more common, researchers are interested in finding out how these digital behaviors may impact the academic life of students. Academic procrastination is one of the major concerns as it the tendency to delay academic task for a prolonged period of time despite being aware of the negative consequences (Ferrari, J. R., Johnson, J. L., & McCown, W. G., 1995). These causes can be attributed to how strongly an individual might be addicted to the usage of social media. Thus, the article highlights the relationship between these two variables. Procrastination is not simply a time management problem but it can be influenced by cognitive, motivational, and environmental factors as well.

The literature review sheds light on the relationship between three variables: Social Media Addiction (Independent Variables), Academic Procrastination (Dependent Variable), and Self-Regulated Learning (Mediating Variable). Social Media Addiction (SMA) refers to the excessive use of social media applications to the point that it disrupts normal routines of life.(C. Andreassen & Pallesen, 2014). Self-regulated learning (SRL) refers to the ability of the individuals who are metacognitively, motivationally, and behaviorally active participants in their own learning process(Zimmerman, 1989). Academic Procrastination (AP) is defined as the voluntary delay of an intended academic course of action despite expecting to be worse off for the delay(Steel, 2007).

The structure of the report is organized as follows: first we have defined the scholarly definitions, dimensions, and measurement approaches for each variable. Next section synthesizes the empirical literature on the relationships among these variables. After that the report highlights the theoretical frameworks underpinning the research model. Then it provides critically evaluation of the existing literature, and then the report is aimed at identifying the specific research gap that motivates the present study. And lastly, this report concludes with a summary of a few key insights.

## **2.2 Explanation of Variables**

### **2.2.1 Definition of Variables**

#### **Social Media Addiction**

When a person uses social networking sites such as Instagram, TikTok, or Facebook excessively enough so that it alters how they conduct their everyday lives, that individual is addicted to social media. For instance, when users of these sites scroll through feeds but don't realize how long they've done so, it becomes clear that there is a problem. Additionally, when people who have social media depend upon checking their notifications because if they don't then, there is anxiety over waiting for them. You'll feel compelled to see your notifications because you want to do so.(Nurmaidah et al., 2025b)

#### **Academic Procrastination**

Academic procrastination entails putting off the completion of your schoolwork or homework until you have completed your other obligations, and saying to yourself "I will do this later," knowing that by doing so you will create stress for yourself and/or receive lower grades, etc. Academic procrastinators experience a multitude of different reasons for procrastinating; however, one of the most common reasons is having difficulty distinguishing between things that they know they should be doing, versus things that they simply would prefer to be doing.(Nurmaidah et al., 2025b)

#### **Self-Regulated Learning (SRL)**

SRL is an approach in which you take responsibility for your learning by managing how you accomplish your tasks (planning/organizing), monitoring the progress of your tasks (tracking), and modifying your study/learning behaviors (adjusting) based on the results of your efforts. In a sense, it is like being coached by yourself; establishing a set of goals and evaluating your success in achieving those goals while being open to attempting different strategies if one does not work for you.(Zimmerman, 2002)



In this study:

- **Social Media Addiction = Independent Variable (X)**
- **Academic Procrastination = Dependent Variable (Y)**
- **Self-Regulated Learning = Mediating Variable (Z)**

### 2.2.2 Dimensions of Variables

Researchers have identified several dimensions when looking at **social media addiction**:

- **Salience** – constantly thinking about using social media
- **Tolerance** – the need to spend greater amounts of time online to achieve satisfaction
- **Withdrawal** – restlessness or feelings of distress when not using social media
- **Modification** – altering one's behavior to use social media
- **Conflict** – social media creating difficulty in the real world
- **Relapse** – resuming previous pattern of behavior after an attempt to decrease use of social media

The dimensions of **academic procrastination** include:

- Delay in the initiation of tasks – delaying the first step in completing a task
- Delay in completion of tasks – excessive time used to complete a task
- Inconstancy of use of time – not using time consistently or using very limited time
- Engagement in activities not related to academic work – using or engaging in most or all of one's time doing something other than working on academic tasks.

### Self-Regulated Learning (SRL)

**SRL** has been broken down into three dimensions:

- Metacognition – thinking about one's own thinking
- Motivation – wanting to do the tasks that are in front of you
- Behavioral control – managing your time and managing your actions

These dimensions assist the researcher in examining each variable in-depth, thus enabling a more thorough analysis than what would have been achieved with only a surface evaluation of each variable.(Tang & He, 2025)

### 2.2.3 Measurement from Past Studies

Most studies use a questionnaire with a 5-point numerical scale from "strongly disagree" to "strongly agree" to measure behaviors of a participant associated with each of these variables.

- Measurement of **social media addiction** includes the frequency of social media use, attachment to social media and/or inability to control their social media use.
- Measurement of **academic procrastination** includes the degree of delay prior to starting academic work and completion of academic work (known as task avoidance).
- **Self-Regulated Learning** - is measured via a combination of thinking strategies, motivational factors, and voluntary behaviors toward developing self-control.

In order for researchers to analyze data/literature using advanced statistical techniques like ***Structural Equation Modelling (SEM)*** and ***Partial Least Squares SEM (PLS-SEM)***, the researcher will have to have access to various types of statistical software or systems which provide them with specific statistical tools needed, i.e. identification of relevant variables; creating a measurement model that is used to determine validity and reliability of the variables and to create an overall measurement model that demonstrates how well the model fits to the data collected from the participants in a study.(C. S. Andreassen et al., 2017)

## 2.3 Review of Empirical Literature

Many studies show that social media addiction is linked with academic procrastination. (Nadarajan et al., 2023a) found that students who use social media too much often delay their academic work. This means that higher social media use increases the chance of procrastination.

The relation between social media addiction and self-regulated learning is also important. ((PDF) *THE RELATIONSHIP BETWEEN TIME MANAGEMENT, SELF REGULATION, AND SMARTPHONE ADDICTION*, n.d.) found that students addicted to social media usually have weak self-regulated learning. This shows that when students cannot control their social media use, they also fail to control their study habits

Research also shows that self-regulated learning reduces academic procrastination. (AquamarineOpenRabbit & Atlas, 2025) found that students with strong self-regulated learning skills are less likely to delay assignments. Good planning and time management help them finish tasks on time.

Finally, some studies explain the mediating role of self-regulated learning. (Karakaya & Altinsoy, 2023) found that self-regulated learning can reduce the negative impact of social media addiction on procrastination. Students who manage their learning well are less affected by social media distractions.

## **2.4 Theoretical Framework**

### **2.4.1 Social Cognitive Theory (SCT)**

When we talk about relationships, we are referring to Social Cognitive Theory (SCT) (Schunk, 2012). This theory explains human behavior as a product of learning, observation, and internal control over behavior through motivation. It views human behavior in relation to the behaviors of others and its relation to the environment in which it takes place.

Previous application of this theory has found that it can be used to understand how students self-regulate their learning and behaviour. Specifically, it has been found that when students are able to manage their behaviour and environment, they are less likely to get distracted by social media and consequently less likely to engage in procrastination behaviours.

This study, drawing on Social Cognitive Theory, investigates the relationship between social media addiction and academic procrastination through students' self-regulated learning. Students with higher self-regulation are able to control distractions and complete tasks on time whereas those with lower self-regulated learning are likely to engage in procrastination.

### **2.4.2 Self-Determination Theory (SDT)**

Self-Determination Theory (SDT) (Ryan & Deci, 2023). This theory examines human motivation and looks at why people are more productive when they are internally motivated, when they have control over things that matter to them, and when their basic psychological needs are met.

In addition to explaining why students are motivated to engage in learning, the theory can also be applied to explain why students engage in academic tasks and avoid academic tasks. Students high in intrinsic motivation are likely to be more engaged in academic tasks and therefore less likely to engage in procrastination. In contrast, students with less intrinsic motivation may be more engaged in procrastination tasks (such as watching TV or posting on social media).

Motivation as described by Self-Determination Theory (Deci & Ryan, 2000) is examined as a predictor of social media addiction and procrastination, and as a predictor of self-regulated

learning. Results indicated that lower motivation was associated with greater social media addiction and procrastination, whereas greater internal motivation was associated with greater self-regulated learning and less procrastination.

### **2.4.3 Integration with the Research Model**

Using Social Cognitive Theory and Self-Determination Theory as a framework for this research is appropriate because the two theories together explain one component of behavior that students use for self-regulated learning (Z) and another component of behavior that students engage in because they are motivated to do so.

The justification for the research model used in this research is further supported by previous research and theoretical approaches. Exposure to social media addiction (X) takes choice and motivation out of the learning process for students. Reduced self-regulated learning, which results in poor time management. Poor self-regulation increases academic procrastination .

The aim of this paper is to investigate potential correlation between social media addiction and procrastination, focusing on the mediating role of self-regulated learning. In addition to the found correlation between SMA and procrastination, as well as the correlation between SMA and self-regulated learning, previous studies found that there is also a correlation between SMA and the current study's variable of interest: SMA decreases procrastination but reduces self-regulated learning.

## **2.5 Critical Evaluation of Literature**

Even though existing literature give significant insights, it is still constrained by several limitations exist:

### **2.5.1 Contextual Limitations**

Most studies are done in certain areas, like Indonesia, which makes it hard to apply the results to other groups.

### **2.5.2 Issues in Methodologies**

This research relies heavily on cross sectional designs  
It also depends on self-reported data, which may introduce bias

### **2.5.3 Limited Range of Variables**

A lot of research only looks at social media and self-regulation, ignoring other factors such as:

- Mental health
- Stress from school/university
- Character traits

### **2.5.4 Findings that don't match**

Some studies show varying strengths of relationships, especially regarding the mediating role of self-regulated learning.

## **2.6 Identification of Research Gap**

While past researchers have sought to understand the relationship between social media addiction, self-regulated learning, and academic procrastination, there remain some gaps in the literature that this study intends to fill.

### **2.6.1 Contextual Gap**

Most of the literature on academic procrastination and social media addiction has been based on university students and high school students in general (Usop & Astuti, 2022; Latipah et al., 2021). On the other hand, vocational high school students have received less attention in previous studies despite having a different learning environment and different learning structures compared to other high school students. This study is intended to bridge this gap in knowledge and understanding by focusing on Office Management and Business Services (MPLB) vocational high school students.

### **2.6.2 Conceptual Gap**

Past research has identified self-regulated learning as a moderating variable for the relationship between social media use and academic procrastination (Yaningsih & Fachrurrozie, 2019). However, the specific role of self-regulated learning as a mediating variable, i.e., the process by which social media addiction leads to academic procrastination, is not sufficiently explored. It is crucial to understand the mediation process as it not only shows the existence of the relationship but also explains the mechanism of the relationship.

### **2.6.3 Methodological Gap**

A significant proportion of the past studies has been based on basic correlation and regression analyses. This has restricted the depth of the analytical approach in the presence of more than one variable (Latipah et al., 2021; Rizkyani et al., 2021). In the current study, the analytical approach used is Structural Equation Modeling – Partial Least Squares (SEM-PLS), which helps to examine the direct and indirect impacts simultaneously.

### **2.6.4 Significance of Bridging This Gap**

The importance of bridging this gap lies in the fact that vocational students form a growing and under-represented group in educational research. The academic environment in which vocational students learn is unique in its nature. This means that if researchers have already conducted studies in academic settings like universities and general high schools, the results cannot be entirely applicable to vocational students. This study will make significant new knowledge by using a rigorous statistical approach to examine the role of self-regulated learning as a mediator in bridging the gap in academic procrastination in vocational schools.

## **2.7 Conclusion**

This literature review shows that how social media addiction actually contributes to academic procrastination and how self-regulated learning serves a very protective and mediating function. Theoretical frameworks such as Self-Determination Theory and Cognitive Behavioral Theory are very important in these relationships. However, limitations in context, methodology, and conceptual show us a distinct research gap, especially within vocational education setting. Future research must address these all these gaps to get a more thorough understanding of student learning behavior.



## **CHAPTER 3: RESEARCH DESIGN**

### **3.1 Introduction**

This report presents a detailed account of the research process adopted to investigate the central questions of this study. It describes the methodological choices made at each stage from the overall research design to data collection and analysis and justifies why these choices are appropriate for addressing the study's objectives. The chapter is structured to first establish the research design, then describe the target population and sampling strategy, followed by an explanation of how variables were operationalized and measured, how data were gathered, and finally, how the data were analyzed to test the hypotheses.

### **3.2 Population and Sample**

#### **3.2.1 Target Population**

The population of this research was students in X and XI class of Office Management Business Services (MPLB) program of SMK Negeri 1 Bogor, Indonesia. The population used because the variables in this research were fit with this population. Vocational high school student in Indonesia are the people who use social media intensively and addictively. They usually look at social media every day for hours. Students tend to be more easily affected by the academic pressure provided by assignments, deadlines and assessments. So, they are easier to have academic procrastination behavior. Hence, it is appropriate for social media addictive usage to be studied in this population.

#### **3.2.2 Sample Size**

In this study, the sample consisted of 134 students of the population of 206 students. A sampling size was calculated by using Isaac and Michael formula with error rate of 5%. When using this calculation, 134 respondents are obtained and will constitute an adequate sample of the population. This sample is also appropriate for a SEM-PLS analysis. SEM-PLS is best suited for a small to medium sample.

#### **3.2.3 Sampling Technique**

The sampling of this study used a proportionate stratified random sampling. The population studied was categorized into subjects' determination with the method of stratified based on class groups (grade X and XI MPLB). Then, the samples of population were taken randomly based on determined to proportional of the subjects by multiple-stage.

### 3.2.4 Inclusion and Exclusion Criteria

| Criteria          | Included                                                              | Excluded                                                                  |
|-------------------|-----------------------------------------------------------------------|---------------------------------------------------------------------------|
| Enrollment Status | currently enroll students pursuing a Grade X or XI MPLB program       | Students from other programs/grades                                       |
| Social Media Use  | Active users (Students who stay constantly connected to social media) | Non-users (Students who seldom use social media)                          |
| Academic Status   | studying on a regular, full schedule                                  | Partial-year courses Part-time or visiting students                       |
| Consent           | willing to participate by choice                                      | Students who possibly do not agree to participate or can retrieve consent |
| Availability      | Available on campus during data collection                            | Students away on leave or internship or exchange                          |

### 3.3 Unit of Analysis

In this study, the unit of analysis is the individual student. It shows that the research doesn't look at the whole class or school together. It focuses on each student one by one.

The study checks three main things for every student:

1. Social media addiction for the students (how much they use social media and how it affects their study)
2. Their self-regulated learning (how well they can plan control and manage their study)
3. Their academic procrastination (how often they delay or postpone their school work).

By choosing the student as the unit of analysis, the research can measure personal behavior and thinking more clearly. It shows the direct link between social media use, study, habits, and procrastination at the individual level instead of mixing results from groups or organizations.

### **3.4 Proposed Data Collection Procedure**

The data for this study was collected through an organized and systematic methodology employing a self-administered questionnaire. To make sure everything was accurate, reliable, and that the rules were followed, the method had several different steps.

First of all we performed a **preliminary study** by giving a questionnaire to 30 participants to find the presence of social media addiction and academic procrastination in students. This step helped us in confirming the relevance of the research problem.

Then the **ethical considerations** were kept in mind. All the participants were told that what was the main purpose of the study before it took place. Their involvement was voluntary, consensual, and confidentiality of their identity and their answers was promised to maintain trust and honesty.

Thirdly, the **questionnaire was given** to the selected sample of all the 134 students of different classes using a proportional approach. The researcher made sure that all respondents clearly understood all the questions fully and help was given whenever it was needed.

Fourth, **data collection** was completed by gathering the filled questionnaires immediately after completion. This helped minimize missing data and ensured a high response rate.

Finally, the collected data underwent **data entry and cleaning**, where responses were organized, checked for completeness, and prepared for further statistical analysis.

### **3.5 Development of Data Collection Instrument**

The main data collection tool in this project is a structured questionnaire. It is a five-point Likert scale, where 1 means strongly disagree and 5 means strongly agree. The questionnaire that is listed below was designed to measure the three main variables on which the main research is based on.

Before collecting the data, the questionnaire was tested to make sure that it is accurate and reliable.

- Construct validity (AVE values above 0.50): The questionnaire properly measures what it suppose to measure.
- **Discriminant validity (HTMT values below 0.90): Each variable that is tested in the research is different from the other variables.**
- Reliability (Cronbach's Alpha and Composite Reliability above 0.90): The questionnaire is consistent and accurate as it gives stable and dependable results.

**Table:** (Nurmaidah et al., 2025c)

Table 4. Construct Reliability and Validity

| Variable                     | Cronbach's Alpha | Composite Reliability (rho_a) | Composite Reliability (rho_c) | AVE   |
|------------------------------|------------------|-------------------------------|-------------------------------|-------|
| Social Media Addiction (X)   | 0,946            | 0,974                         | 0,950                         | 0,574 |
| Academic Procrastination (Y) | 0,937            | 0,942                         | 0,946                         | 0,616 |
| Self-Regulated Learning (Z)  | 0,933            | 0,939                         | 0,942                         | 0,598 |

---

## QUESTIONNAIRE

Dear Participant,

Our team is conducting research on Social Media Addiction, Self-Regulated Learning, and Academic Procrastination among students.

This study aims to examine how social media addiction affects academic procrastination and how self-regulated learning plays a mediating role. Your responses will remain confidential and will be used only for academic purposes.

### Personal Information

1. How old are you? (in years) \_\_\_\_\_
2. What is your grade level?
  - a. Grade X
  - b. Grade XI

3. What is your gender?

- a. Male
- b. Female

4. How many hours do you spend on social media daily?

- a. Less than 2 hours
- b. 2–3 hours
- c. 3–6 hours
- d. More than 6 hours

### Section 1: Social Media Addiction

| Code | Statement                                                     | Strongly disagree | Disagree | Neither | Agree | Strongly agree |
|------|---------------------------------------------------------------|-------------------|----------|---------|-------|----------------|
| SMA1 | I delay starting my academic tasks because I use social media | 1                 | 2        | 3       | 4     | 5              |
| SMA2 | I spend more time on social media than I initially planned    | 1                 | 2        | 3       | 4     | 5              |
| SMA3 | I find it difficult to reduce my time on social media         | 1                 | 2        | 3       | 4     | 5              |
| SMA4 | I feel distracted from my studies due to social media use     | 1                 | 2        | 3       | 4     | 5              |
| SMA5 | I prefer using social media over completing academic tasks    | 1                 | 2        | 3       | 4     | 5              |

### Section 2: Self-Regulated Learning

| Code | Statement                                             | Strongly disagree | Disagree | Neither | Agree | Strongly agree |
|------|-------------------------------------------------------|-------------------|----------|---------|-------|----------------|
| SRL1 | I plan my study activities before starting my work    | 1                 | 2        | 3       | 4     | 5              |
| SRL2 | I set goals for my academic tasks                     | 1                 | 2        | 3       | 4     | 5              |
| SRL3 | I can control myself from distractions while studying | 1                 | 2        | 3       | 4     | 5              |

|      |                                                          |   |   |   |   |   |
|------|----------------------------------------------------------|---|---|---|---|---|
| SRL4 | I manage my time effectively when completing assignments | 1 | 2 | 3 | 4 | 5 |
| SRL5 | I stay motivated to complete my academic tasks on time   | 1 | 2 | 3 | 4 | 5 |

### Section 3: Academic Procrastination

| Code | Statement                                                     | Strongly disagree | Disagree | Neither | Agree | Strongly agree |
|------|---------------------------------------------------------------|-------------------|----------|---------|-------|----------------|
| AP1  | I delay completing my academic assignments until the deadline | 1                 | 2        | 3       | 4     | 5              |
| AP2  | I find it difficult to start my academic tasks                | 1                 | 2        | 3       | 4     | 5              |
| AP3  | I feel stressed because I postpone my assignments             | 1                 | 2        | 3       | 4     | 5              |
| AP4  | I often choose to do other activities instead of studying     | 1                 | 2        | 3       | 4     | 5              |
| AP5  | I rush to complete assignments at the last minute             | 1                 | 2        | 3       | 4     | 5              |

*Note: The above questionnaire is generated by our group members and the actual questionnaire which is tested is not publicly available.*

### 3.8 Proposed Data Analysis Techniques

Analysis of the data was done through Structural Equation Modeling by Partial Least Squares (SEM-PLS), using SmartPLS 4.1. This method was selected due to its suitability in analyzing mediating relationships with numerous latent constructs and being suitable for the number of participants in the current study. The process took three steps:

**Measurement Model (Outer Model):** The following approach was adopted to test for reliability and validity of the measures. Reliability of the construct was measured using Cronbach's Alpha and Composite Reliability and the ideal value was expected to surpass 0.90. Validity of the construct was determined through AVE measurement and the acceptable value was above 0.50. Discriminant validity of the measures was confirmed using HTMT and the required value must fall below 0.90.

**Structural Model (Inner Model):** Structural relationships in the structural model were evaluated based on path coefficient, which indicates the relationship between variables; R-squared that represents how well the dependent variable is predicted; and F-squared representing magnitude of impact.

**Hypothesis Testing:** The influence of both direct and mediation effect was tested using bootstrap analysis. The level of statistical significance used was where T value exceeds 1.96 and p-value is less than 0.05, consistent with common practice within the social sciences. This model enabled the examination of:

- (1)- the direct effect of social media addiction on academic procrastination
- (2)- the effect of social media addiction on self-regulated learning
- (3)- the effect of self-regulated learning on academic procrastination
- (4)- the mediation effect of self-regulated learning between social media addiction and academic procrastination.

### **3.9 Conclusion**

This report has provided a comprehensive account of the research methodology employed in this study. A quantitative, cross-sectional survey design was adopted, drawing a sample of 134 students from SMKN 1 Bogor through proportional stratified random sampling. Data was collected from students via a structured questionnaire using a five point Likert Scale. Analysis was performed using SEM-PLS in SmartPLS 4.1, this followed three stages: Firstly, in order to confirm the reliability, the outer model was evaluated. Secondly, R-squared values and F-squared values were used for assessing the structural model,  $R^2$  of 0.728, indicated a strong explanatory power and F-squared values for individual effect sizes. Thirdly, hypothesis testing examined both direct effects (H1, H2, H3) using path coefficients, T-test, and P-values, and the indirect mediation effect (H4) through bootstrapped Specific Indirect Effect analysis.

The study is transparent and rigorous as it establishes a clear sequence of steps from population from population definition and sampling through instrument development and statistical analysis, drawing meaningful conclusions about the role of social media addiction and self-regulated learning in influencing students' academic procrastination behavior.

## CHAPTER 4: DATA ANALYSIS

### 4.1 Introduction

This chapter provides a general analysis of the data collected from 160 respondents through a structured questionnaire survey. The aims, questions and hypotheses of the study have a large impact on the analysis of the study. The present study examines the relationship between academic procrastination and social media addiction with self-regulated learning as a mediating variable. The chapter has been divided into sections such as data preparation, descriptive statistics, measurement model assessment, structural model evaluation and hypothesis testing.

All analyses in the procedure were performed using the SmartPLS 4.1 software. The method used is Structural Equation Modelling-Partial Least Squares (SEM-PLS).

### 4.2 Data Preparation

Before any analysis, the dataset was looked over pretty carefully, and then was set up in a way that kept the data quality steady. A bunch of steps were done for that, for example:

- **Data Entry and Coding:**

All the answers from the questionnaire got entered into a spreadsheet , and then they were coded numerically too. The Likert scale items, those 1 to 5 responses were kept as plain integers rather than anything else. For demographics like gender and enrollment level, those weren't treated as numbers, instead they turned into categorical text fields.

- **Missing Value Treatment:**

The data set was screened for any missing values. But, well, not much significant missing data showed up , because the questionnaire felt pretty straightforward , and all questions had to be answered. Also, the response rate came out to 100%, so there wasn't really any slack.

- **Data Import:**

The cleaned dataset got imported into SmartPLS 4.1, in CSV format, so basically it was loaded that way. Then the measurement model was built by linking every indicator SMA1–SMA5, SRL1–SRL5, AP1–AP5 to its own latent construct, you could say.



- **Normality Screening:**

For each indicator and every latent variable (LV) score, descriptive statistics were computed including skewness and excess kurtosis, in that way we get some view. Since PLS-SEM is kind of non-parametric, a strict normality assumption is not really required, yet normality screening was still performed, just to better describe the overall data distribution.

## 4.3 Descriptive Statistics

### 4.3.1 Respondent Profile

The study was conducted and it collected responses from 160 participants as shown in the table below

**Table 4.1: Respondent Profile by Enrollment Level**

| Enrollment Level | Frequency  | Percentage (%) |
|------------------|------------|----------------|
| School           | 46         | 28.8           |
| College          | 55         | 34.4           |
| University       | 59         | 36.9           |
| <b>Total</b>     | <b>160</b> | <b>100.0</b>   |

The majority of respondents were from the university going student about 36.9% as we can see in the above table.

**Table 4.2: Respondent Profile by Gender**

| Gender | Frequency | Percentage (%) |
|--------|-----------|----------------|
| Female | 112       | 70.0           |

|              |            |              |
|--------------|------------|--------------|
| Male         | 48         | 30.0         |
| <b>Total</b> | <b>160</b> | <b>100.0</b> |

As shown in the table 4.2 the most respondents were female students that are about 70% and 30% are filled from male students.

**Table 4.3: Respondent Profile by Daily Social Media Usage**

| Daily Social Media Usage | Frequency  | Percentage (%) |
|--------------------------|------------|----------------|
| Less than 2 hours        | 5          | 3.1            |
| 2–3 hours                | 21         | 13.1           |
| 3–6 hours                | 84         | 52.5           |
| More than 6 hours        | 50         | 31.3           |
| <b>Total</b>             | <b>160</b> | <b>100.0</b>   |

Table 4.3 reveals that 52.5% of respondents spend 3–6 hours on social media daily, while a further 31.3% spend more than 6 hours. This reveals that the 83.8% students exceed the 2–3 hour daily limit.

#### 4.3.2 Descriptive Statistics of Study Variables

The table 4.4 shows the statistical analysis of three variables used in the study. This statistical report is generated from the analysis of Smart PLS .

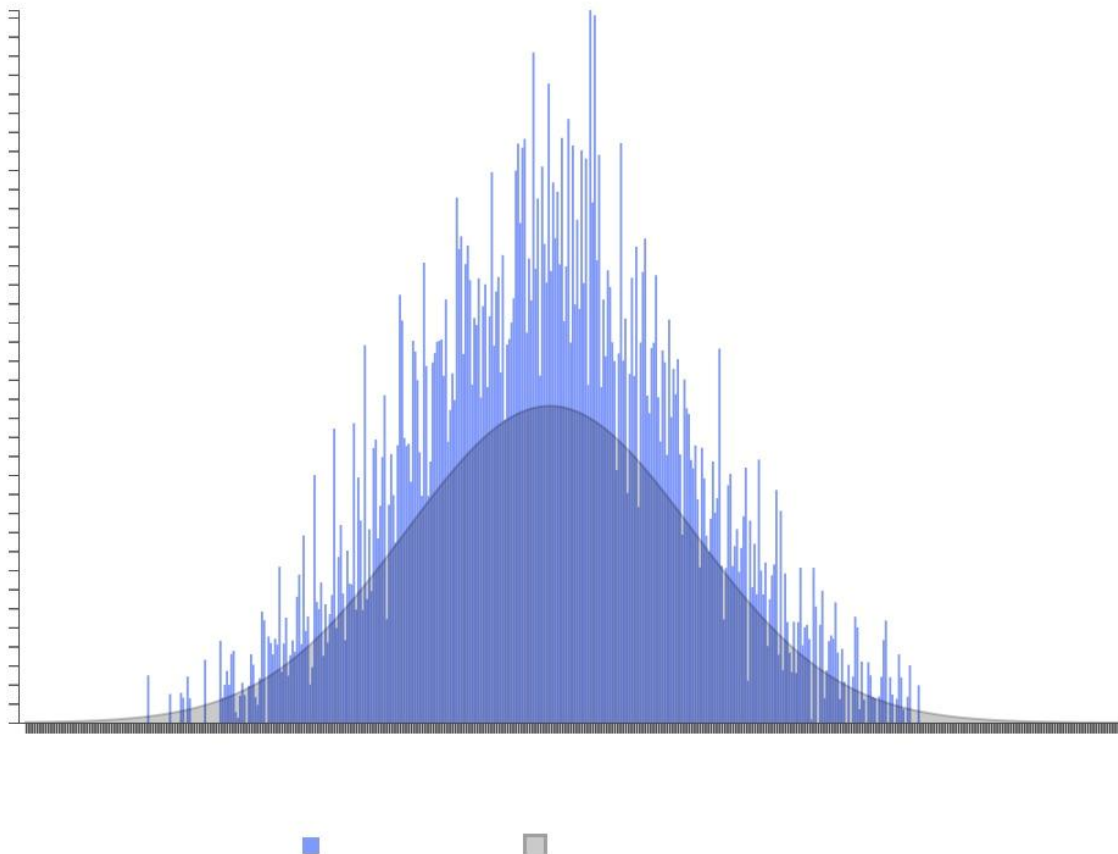
**Table 4.4: Descriptive Statistics of Latent Variables and Indicators**

| Variable / Item               | Mean  | Median | Obs. Min | Obs. Max | Std. Dev. | Skewness |
|-------------------------------|-------|--------|----------|----------|-----------|----------|
| LV — Social Media Addiction   | 3.581 | 3.6    | 1.0      | 5.0      | 0.666     | -0.508   |
| LV — Self-Regulated Learning  | 3.283 | 3.2    | 1.8      | 5.0      | 0.533     | -0.133   |
| LV — Academic Procrastination | 3.454 | 3.4    | 1.6      | 5.0      | 0.662     | -0.182   |
| SMA1                          | 3.688 | 4.0    | 1        | 5        | 1.174     | -0.775   |
| SMA2                          | 3.612 | 4.0    | 1        | 5        | 1.124     | -0.925   |
| SMA3                          | 3.631 | 4.0    | 1        | 5        | 1.176     | -0.552   |
| SMA4                          | 3.731 | 4.0    | 1        | 5        | 0.979     | -0.646   |
| SMA5                          | 3.244 | 3.0    | 1        | 5        | 1.117     | -0.305   |
| SRL1                          | 3.494 | 4.0    | 1        | 5        | 0.949     | -1.312   |
| SRL2                          | 3.531 | 4.0    | 1        | 5        | 0.974     | -1.194   |
| SRL3                          | 2.894 | 3.0    | 1        | 5        | 1.046     | -0.182   |
| SRL4                          | 3.188 | 3.0    | 1        | 5        | 1.020     | -0.170   |
| SRL5                          | 3.306 | 4.0    | 1        | 5        | 1.067     | -0.857   |
| AP1                           | 3.500 | 4.0    | 1        | 5        | 1.157     | -0.477   |
| AP2                           | 3.538 | 4.0    | 1        | 5        | 0.914     | -0.533   |

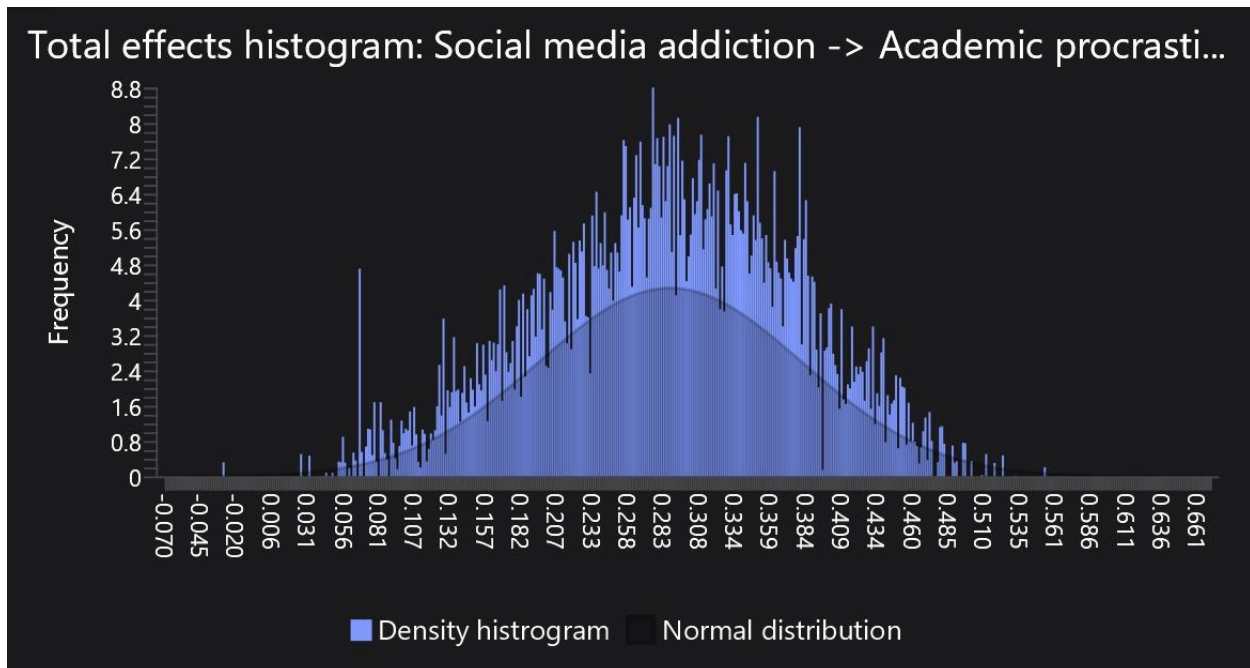
|     |       |     |   |   |       |        |
|-----|-------|-----|---|---|-------|--------|
| AP3 | 3.225 | 4.0 | 1 | 5 | 1.245 | -0.298 |
| AP4 | 3.569 | 4.0 | 1 | 5 | 1.004 | -1.032 |
| AP5 | 3.438 | 4.0 | 1 | 5 | 1.278 | -0.449 |

The mean LV score for Social Media Addiction ( $M = 3.581$ ) indicates the high level of addiction in the sample. The Self-Regulated Learning LV score ( $M = 3.283$ ) is lower and it suggests that students have lower control on their learning. The Academic Procrastination LV score ( $M = 3.454$ ) is above average and it shows that delaying tasks are common among the students. All skewness values are negative which means that the responses are highly side of scale.

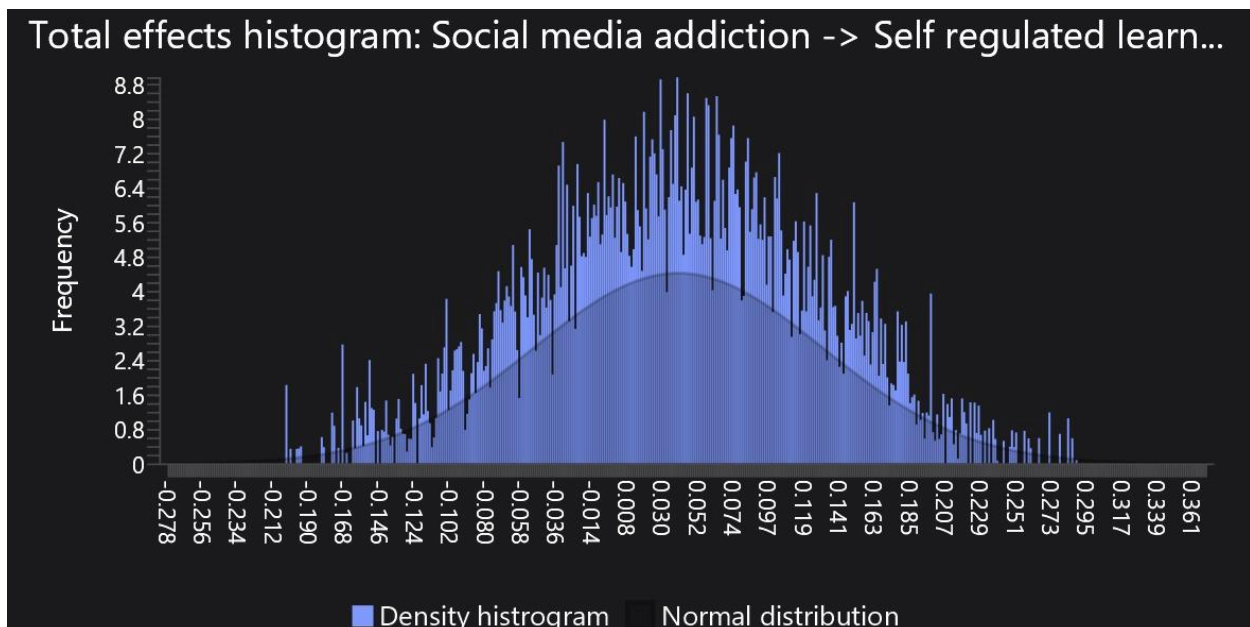
### 4.3.3 Visualizations using Histogram

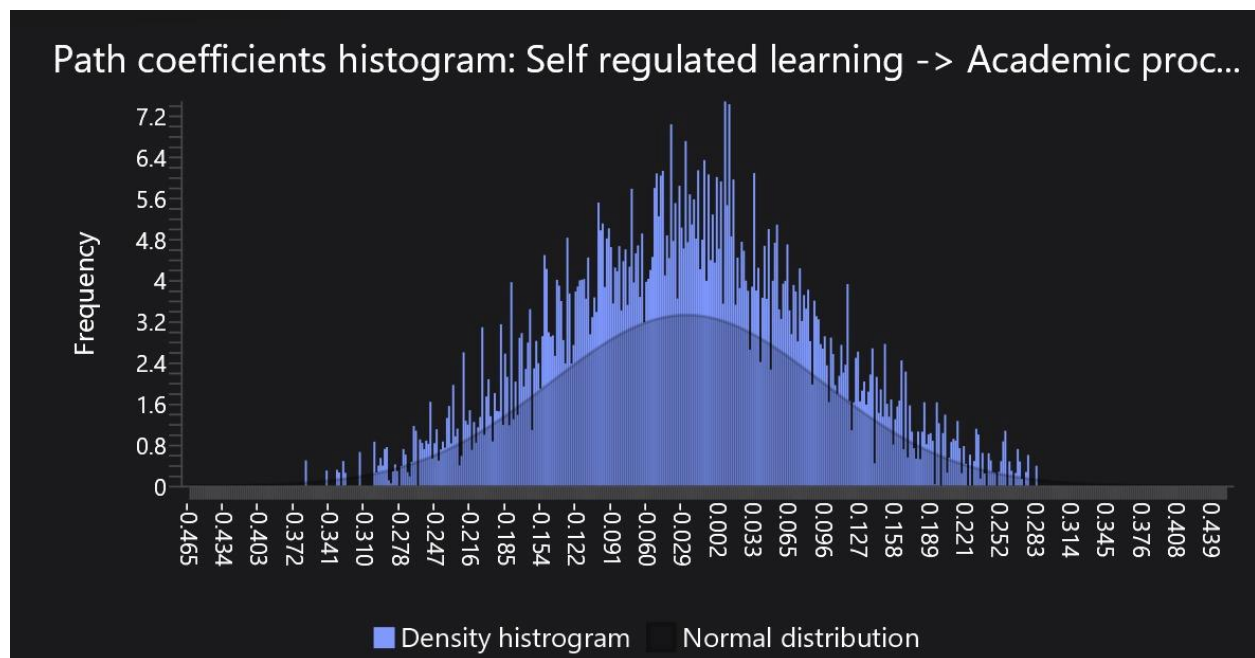


**(i) SMA → AP**



**(ii) SMA → SRL**



**(iii) SRL → AP****4.4 Measurement Model Evaluation (Outer Model)**

Before checking the relationship of variables, the measurement model was tested to make sure all three constructs are reliable and valid. This involved checking internal consistency reliability, convergent validity and discriminant validity.

**4.4.1 Construct Reliability****Table 4.5: Construct Reliability**

| Construct                     | Cronbach's Alpha | Composite Reliability (rho_c) | AVE   |
|-------------------------------|------------------|-------------------------------|-------|
| Social Media Addiction (SMA)  | 0.946            | 0.950                         | 0.574 |
| Self-Regulated Learning (SRL) | 0.933            | 0.942                         | 0.598 |
| Academic Procrastination (AP) | 0.937            | 0.946                         | 0.616 |

As shown in Table 4.5, all Cronbach's Alpha values are above 0.90, which shows very strong internal consistency among the items of each construct. The Composite Reliability values are also above 0.90, confirming that the indicators reliably measure their related constructs. The Average Variance Extracted (AVE) values range from 0.574 to 0.616, which are higher than the required value of 0.50. This shows good convergent validity, meaning each construct explains more than half of the variation in its indicators.

#### 4.4.2 Discriminant Validity (HTMT)

| Construct Pair                                     | HTMT Value |
|----------------------------------------------------|------------|
| Academic Procrastination ↔ Social Media Addiction  | 0.523      |
| Self-Regulated Learning ↔ Social Media Addiction   | 0.260      |
| Self-Regulated Learning ↔ Academic Procrastination | 0.787      |

All HTMT values in Table 4.6 are below the recommended limit of 0.90. The lowest value is between Self-Regulated Learning and Social Media Addiction (0.260), while the highest value is between Self-Regulated Learning and Academic Procrastination (0.787), which is still below the required threshold. These results show that the three constructs are clearly different from each other and have good discriminant validity.

#### 4.4.3 Collinearity Assessment (VIF)

The VIF values were calculated to check whether the collinearity exists in the model or not. The VIF values ranged from 1.000 to 1.069, which are below the accepted limit of 5.0. This shows us that all constructs are independent from each other.

## 4.5 Structural Model Evaluation (Inner Model)

### 4.5.1 R-Square and Predictive Power

The percentage of variance in the endogenous (dependent and mediating) variables explained by the model's predictors is shown by the R-squared ( $R^2$ ) values.

**Table 4.7: R-Square Values**

| Construct                     | $R^2$ | $R^2$ Adjusted |
|-------------------------------|-------|----------------|
| Academic Procrastination (AP) | 0.086 | 0.074          |
| Self-Regulated Learning (SRL) | 0.003 | -0.003         |

The model's  $R^2$  for academic procrastination is 0.086, meaning that the combination of social media addiction and self-regulated learning accounts for 8.6% of the variance in academic procrastination. The  $R^2$  for self-regulated learning is 0.003, indicating that only 0.3% of the variance in self-regulated learning can be explained by social media addiction. In light of the results of the hypothesis test, Chapter 5 goes into additional detail about how modest these values are.

### 4.5.2 Effect Size (F-Square)

The practical relevance of each predictor's contribution to the dependent variable is measured by the F-squared ( $F^2$ ) values.

**Table 4.8: F-Square Effect Sizes**

| Path                                              | $F^2$ | Effect Size  |
|---------------------------------------------------|-------|--------------|
| Social Media Addiction → Academic Procrastination | 0.094 | Small-Medium |



|                                                    |        |            |
|----------------------------------------------------|--------|------------|
| Self-Regulated Learning → Academic Procrastination | 0.0005 | Negligible |
| Social Media Addiction → Self-Regulated Learning   | 0.003  | Negligible |

Social media addiction has a minor-to-medium impact on academic procrastination ( $F^2 = 0.094$ ), according to Cohen's (1988) criteria ( $F^2 > 0.02$  = small;  $\geq 0.15$  = medium;  $\geq 0.35$  = high). In this paradigm, the impacts of social media addiction on self-regulated learning and academic procrastination on self-regulated learning are minimal.

## 4.6 Hypothesis Testing

### 4.6.1 Direct Effects (H1, H2 ,H3)

Using SMART PLS 4.1 the dataset went through a bootstrapping test with 5000 subsamples. The strength relationship is only considered significant if the p-value is less than 0.05 and the t-test value exceeds greater than 1.96.

**Table 4.9: Direct Effects — Path Coefficient Results**

| Hypothesis / Path | $\beta$ (O) | Mean (M) | STDEV | T-stat | P-value | Decision      |
|-------------------|-------------|----------|-------|--------|---------|---------------|
| H1: SMA → AP      | 0.292       | 0.285    | 0.095 | 3.066  | 0.002   | Supported     |
| H2: SMA → SRL     | 0.045       | 0.040    | 0.090 | 0.495  | 0.621   | Not Supported |
| H3: SRL → AP      | -0.025      | -0.026   | 0.120 | 0.211  | 0.833   | Not Supported |

**H1: Social Media Addiction → Academic Procrastination ( $\beta = 0.292$ ,  $T = 3.066$ ,  $p = 0.002$ )**

The hypothesis that SMA positively effects AP is supported the t-test and p-values. Students are more likely to spend more time on smartphones and delay task if they exhibit high levels of social media addiction (Nadarajan et al., 2023b). These previous studies found a similar positive relationship between SMA and AP.

**H2: Social Media Addiction → Self-Regulated Learning ( $\beta = 0.045$ ,  $T = 0.495$ ,  $p = 0.621$ )**

From the above table it is observed that the  $p > 0.05$  and  $T < 1.96$ , which indicates that the relationship between SMA and SRL is not significant. This implies that in this dataset SMA does not have direct impact on SRL capacity of students. This result may differ from the original article (Nurmaidah et al., 2025a). In this research SMA was found to have negative impact on SRL of students, meaning it reduces the ability of students to learn new things.

**H3: Self-Regulated Learning → Academic Procrastination ( $\beta = -0.025$ ,  $T = 0.211$ ,  $p = 0.833$ )**

The table highlights a negative correlation, which indicates that students with a higher SRL capacity are not that affected by AP, however the strength of the relation not significant i.e.  $p > 0.05$  and  $T < 1.96$ . In this sample, students having a high SRL still does not significantly negate of AP when controlling for SMA.

#### 4.6.2 Indirect Effect: Mediation Test (H4)

The mediation effect of SRL was found using the special indirect effects analysis in the bootstrapping test. The indirect path tested was: Social Media Addiction → Self-Regulated Learning → Academic Procrastination.

**Table 4.10: Specific Indirect Effect — H4 Mediation Test**

| Indirect Path | $\beta$ (O) | Mean (M) | STDEV | T-stat | P-value | Decision |
|---------------|-------------|----------|-------|--------|---------|----------|
|               |             |          |       |        |         |          |

|                |        |       |       |       |       |               |
|----------------|--------|-------|-------|-------|-------|---------------|
| SMA → SRL → AP | -0.001 | 0.003 | 0.012 | 0.092 | 0.927 | Not Supported |
|----------------|--------|-------|-------|-------|-------|---------------|

#### **H4: Self-Regulated Learning as Mediator ( $\beta = -0.001$ , $T = 0.092$ , $p = 0.927$ )**

The hypothesis of SMA indirectly influencing AP through SRL is not supported as the  $p > 0.05$  and  $T < 1.96$ , which indicated an insignificant relationship. The 95% bias-corrected confidence interval spans between -0.033 and 0.013], which includes zero, further confirming the absence of a significant mediation effect. This result is supported by the previous two hypotheses H2 and H3. Since neither the SMA→SRL nor the SRL→AP paths are significant, the indirect pathway through SRL is also non-significant.

#### **4.6.3 Regression Analysis Results**

Although the original research does not include a regression analysis. The SmartPLS linear regression analysis was additionally run to examine the combined predictive model for Academic Procrastination. Table 4.11 presents the standardized coefficients and ANOVA results.

**Table 4.11: Regression Summary — Dependent Variable: Academic Procrastination**

| Predictor                    | Unstd. $\beta$ | Std. $\beta$ | SE    | T-value | P-value |
|------------------------------|----------------|--------------|-------|---------|---------|
| Social Media Addiction (LV)  | 0.294          | 0.292        | 0.077 | 3.838   | 0.000   |
| Self-Regulated Learning (LV) | 0.071          | 0.057        | 0.095 | 0.748   | 0.456   |
| Intercept                    | 2.332          | —            | 0.413 | 5.646   | 0.000   |

**Table 4.12: ANOVA Summary**

| Source     | Sum of Squares | df  | F-value | P-value |
|------------|----------------|-----|---------|---------|
| Regression | 6.281          | 2   | 7.635   | 0.000   |
| Error      | 64.582         | 157 | —       | —       |
| Total      | 70.864         | 159 | —       | —       |

The ANOVA results confirm that the overall regression model is statistically significant ( $F = 7.635$ ,  $p < 0.001$ ), indicating that the combination of Social Media Addiction and Self-Regulated Learning significantly predicts Academic Procrastination. However, only Social Media Addiction makes a significant independent contribution (Std.  $\beta = 0.292$ ,  $p < 0.001$ ), while Self-Regulated Learning does not (Std.  $\beta = 0.057$ ,  $p = 0.456$ ). The  $R^2$  for this regression model is 0.089, and the Durbin-Watson statistic is 2.284, indicating no autocorrelation in the residuals.

#### 4.7 Summary of Hypothesis Testing

This chapter gives an exhaustive overview of the hypotheses tested in this research. The analysis was performed in SmartPLS 4.1 using bootstrap sampling techniques (5,000 subsamples). If the T-statistic value was higher than 1.96, and the p-value lower than 0.05, the hypothesis was considered accepted. See Table 4.13 below for all findings.

Table 4.13: Summary of Hypothesis Testing Results

| H  | Path                  | $\beta$ (O) | Mean  | T-Stat | p-value | Significant?       | Decision      |
|----|-----------------------|-------------|-------|--------|---------|--------------------|---------------|
| H1 | SMA $\rightarrow$ AP  | 0.292       | 0.285 | 3.066  | 0.002   | Yes ( $p < 0.05$ ) | Supported     |
| H2 | SMA $\rightarrow$ SRL | 0.045       | 0.040 | 0.495  | 0.621   | No ( $p > 0.05$ )  | Not Supported |

| H  | Path                                               | $\beta$ (O) | Mean   | T-Stat | p-value | Significant?      | Decision      |
|----|----------------------------------------------------|-------------|--------|--------|---------|-------------------|---------------|
| H3 | SRL $\rightarrow$ AP                               | -0.025      | -0.026 | 0.211  | 0.833   | No ( $p > 0.05$ ) | Not Supported |
| H4 | SMA $\rightarrow$ SRL $\rightarrow$ AP (Mediation) | -0.001      | 0.003  | 0.092  | 0.927   | No ( $p > 0.05$ ) | Not Supported |

Note: SMA = Social Media Addiction; SRL = Self-Regulated Learning; AP = Academic Procrastination;  $\beta$  (O) = original sample path coefficient; T-Stat = T-statistic from bootstrapping; Significance threshold:  $T > 1.96$ ,  $p < 0.05$ .

**Hypothesis 1 (H1):** Social media addiction positively and significantly impacts academic procrastination ( $\beta = 0.292$ ,  $t = 3.066$ ,  $p = 0.002$ ). The hypothesis holds because  $p < 0.05$  and  $t > 1.96$ . Students who exhibit higher social media addiction behavior tend to procrastinate more in their academics.

**Hypothesis 2 (H2):** Social media addiction does not have any significant impact on self-regulated learning ( $\beta = 0.045$ ,  $t = 0.495$ ,  $p = 0.621$ ). The hypothesis does not hold because  $p > 0.05$ . Although there exists a positive association between social media addiction and self-regulated learning, it contradicts the theoretical expectation of a negative relationship.

**Hypothesis 3 (H3):** Self-regulated learning does not significantly affect academic procrastination ( $\beta = -0.025$ ,  $t = 0.211$ ,  $p = 0.833$ ). Despite having the negative relationship as expected, this hypothesis does not hold because  $p > 0.05$ . The effect is small and insignificant.

**Hypothesis 4 (H4):** Self-regulated learning does not mediate the association between social media addiction and academic procrastination (indirect  $\beta = -0.001$ ,  $t = 0.092$ ,  $p = 0.927$ ). The confidence interval [lower bound = -0.033; upper bound = 0.013] contains zero, suggesting that self-regulated learning fails to mediate the relationship. This hypothesis does not hold since it is expected for the mediator to be significant for any mediating effect to occur.

## Overall

On the whole, among the four hypotheses, only H1 gained acceptance. The relationship between social media addiction and academic procrastination is validated. Self-regulated learning, on the other hand, could not act as the mediator for this group of learners. In sum, it can be suggested that the link between these variables could differ based on the learner's background.

## **CHAPTER 5: DISCUSSION AND CONCLUSION**

### **5.1 Introduction**

This chapter discusses the interpretation of the results that were found in the preceding chapter. The chapter puts forward the research's limitations and as well as the recommendations for future research. Finally, it concludes by summarizing the major findings of the research.

### **5.2 DISCUSSION OF KEY FINDINGS**

#### **5.2.1 H1: Social Media Addiction → Academic Procrastination**

The following section will consider each of the hypotheses separately in light of the current results compared to the results obtained by other authors. When there is agreement between the findings, the reasons for such congruence will be explained. In case of disagreements, plausible reasons will be considered.

#### **Results**

There was sufficient evidence for the hypothesis ( $\beta = 0.292$ ,  $T = 3.066$ ,  $p = 0.002$ ). Thus, social media addiction has a significant positive impact on academic procrastination.

#### **Comparison with the literature**

These results correspond to previous findings in the field. In their study of Indonesian vocational high school students, Nurmaidah et al. (2025) found an equivalent relationship ( $\beta = 0.399$ ,  $p < 0.001$ ). While the effect size in the current research ( $\beta = 0.292$ ) is slightly lower, it might be attributed to the increased diversity of the sample since the current paper involved participants of different ages and educational institutions, whereas the sample of Nurmaidah et al. was limited to vocational students. Additionally, internet addiction was significantly correlated with academic procrastination by Apriany et al. (2023), and a similar relationship was found in Malaysian university students by Nadarajan et al. (2023).

#### **Rationale for the association**

Social media offers instant gratification, which takes various forms, from likes and videos to comments. Academic tasks, on the other hand, provide delayed gratification. If the students' behaviors indicate addiction, they will prefer the former because of its immediacy and fun.

### 5.2.2 H2: Social Media Addiction → Self-Regulated Learning

**Result:** My finding indicated that the hypothesis was not supported ( $\beta = 0.045$ ,  $T = 0.495$ ,  $p = 0.621$ ). Thus, the association was not significant. It should be mentioned that the nature of the relation was positive, in contrast to the expected result.

#### **Contradiction with Existing Studies:**

The findings of this research contradict some existing studies. For instance, in their study Nurmaidah et al. (2025) found a significant negative correlation between social media addiction and self-regulated learning ( $\beta = -0.253$ ,  $p < 0.001$ ), which means that the former was a barrier to the latter. Similar results were obtained in the research by Anggreani (2018) and Azizah & Muslikah (2021).

#### **Possible reasons for difference:**

- **Heterogeneous sample:** The sample in this study included pupils, students, and even university graduates. University students are generally older and might be characterized by strong self-control and therefore would be less likely influenced by social media. However, the sample of the original study was homogenous as it consisted of vocational high school students.
- **Lack of variance:** The scores of self-regulated learning among current sample showed lack of variance ( $M = 3.283$ ,  $SD = 0.533$ ), which means that the participants' scores had little difference.
- **Compartmentalization:** Social media can be used only during free time. Therefore, students might separate their academic and recreational lives.

### 5.2.3 H3: Self-Regulated Learning → Academic Procrastination

#### **Results:**

The predicted effect was unsupported by the findings ( $\beta = -0.025$ ,  $t = 0.211$ ,  $p = 0.833$ ). While consistent with expectations regarding its negative direction, it failed to yield any statistically significant findings due to low magnitudes.

#### **Comparison with existing literature:**



This study differs from previous studies mostly due to the finding itself. According to Rizkyani et al. (2021), self-regulated learning positively correlates with the level of academic procrastination. A similar positive correlation was found by Santika & Sawitri (2016). This trend continues in the work by Abdullah et al. (2024). Notably, the strongest effect of the type has been recorded by Nurmaidah et al. (2025), with a  $\beta$  value of -0.660 ( $p < 0.001$ ). In this case, self-regulated learning served as the driving force for procrastination. In contrast, no evidence for a relationship of this kind can be found in the current study.

#### **Reasons for the inconsistency:**

- **Heterogeneity of the sample:** The studied sample includes representatives from three educational levels, in each of which self-regulation and procrastination can correlate in different ways. Aggregation of results might lead to counterbalancing, and therefore to insignificant findings.
- **Importance of external variables:** An  $R^2$  of 0.086 suggests that there are still many external factors that can contribute to procrastination but were not included in the model.
- **Limited validity of self-regulation measurement:** Self-regulation has only been measured using five questions.

#### **5.2.4 H4: Mediation Effect of Self-Regulated Learning**

Conclusion: The proposed hypothesis could not be supported (indirect  $\beta = -0.001$ ,  $t = 0.092$ ,  $p = 0.927$ ). Since the 95% confidence interval  $[-0.033, 0.013]$  includes zero, this indicates the lack of mediation.

#### **Comparison with the findings of other researchers:**

A number of papers expected SRL to act as a mediator. Thus, Nurmaidah et al. (2025) found the full mediation effect in their study. Özyer and Altınsoy (2023) established self-regulation as the mediator in the relationship between the use of digital media and procrastination in Turkish students, whereas Latipah et al. (2021) identified the mediation effects among Indonesian students.

#### **Reasons for inconsistency with my results:**

Refusal to accept H4 is logical considering the inability to prove H2 and H3. Mediation requires satisfying two conditions:

- The impact of the independent variable (SMA) on the mediator (SRL) should be significant – H2 could not be accepted.
- The impact of the mediator (SRL) on the dependent variable (AP) should be significant – H3 could not be accepted.

As there were no significant impacts, the mediation effect could not occur. In the meantime, while the mediation pathway might appear in the study with more homogeneous student population, it will be absent in a heterogonous group of students. Future research might take other possible mediators into account, including academic motivation, emotion regulation, and digital self-efficacy.

*Table 5.1: Summary of Hypothesis Findings vs. Existing Literature*

| Hypothesis    | Finding       | Matches Literature? | Key Reason for Difference (if any)                                                                    |
|---------------|---------------|---------------------|-------------------------------------------------------------------------------------------------------|
| H1: SMA → AP  | Supported     | Yes                 | Consistent with Nurmaidah et al. (2025), Apriany et al. (2023), Nadarajan et al. (2023)               |
| H2: SMA → SRL | Not Supported | No                  | Heterogeneous sample; older students have stronger self-regulation; limited score variance            |
| H3: SRL → AP  | Not Supported | No                  | Mixed educational levels cancel effects; low R <sup>2</sup> (8.6%); other unmeasured factors dominate |
| H4: Mediation | Not Supported | No                  | Logical result of H2 and H3 both failing; no significant paths to form an indirect effect             |

**Conclusion:** Out of all the hypotheses tested, it can be concluded that only hypothesis one gained acceptance. The data shows that social media addiction has a significant and explicit impact on increasing the levels of academic procrastination in students, which is consistent with

past research. Self-regulated learning, on the other hand, was unable to act as a mediating variable between social media addiction and academic procrastination in this study. Hypotheses two through four were found to be false, most probably because of the diverse sample size, limited variability in the self-regulated learning variable, and lack of other relevant variables in the model.

## **5.3 Implications**

### **5.3.1 Theoretical Implications**

This study adds value to the growing literature in research on topic digital behavior and students' academic outcomes. By partially replicating and expanding the work of Nur Maidah et al., (2025) in a larger, more general sample of students. The research findings support Hypothesis 1 (H1) which represents that social media addiction is a positive effect on academic procrastination. That means the student actively using social media directly facing delays in academic work and procrastinating on academic assignments. However, the H2, H3, and H4 results are not supported and backed by the findings of our research. The findings of our research show that the relationship between Social Media Addiction and Academic Procrastination was not mediated by Self-Regulated Learning. The results indicate that the effect and mediation of self-regulated learning depend on the context, environment and the type of students being studied in the research. SRL behaves differently in different contexts, environments and educational settings.

### **5.3.2 Practical Implications**

In this study, only one hypothesis is supported by the results of the research. Partially supported hypothesis H1, the positive relationship between social media addiction and academic procrastination has evident practical implications. There are still practical implications for educational institutions, teachers and students. Educational institutions need to realize that offering measures and strategies to help students cope with the usage of social media when studying is not to be avoided. Workshops on being digitally healthy, to limit distractions during study, tutoring hours for time management and enhancing self-control and academic self-discipline guidance should be introduced by HEI's. Teachers and college instructors should raise awareness during lessons using self-awareness activities.

## **5.4 Limitations of the Study**

There are several limitations that should be acknowledged when interpreting the findings of this study.

### **Self-Reported Data**

In this study, the self-report questionnaire is used to collect data from the students. Due to this there is a possibility of biasness involving social desirability. Participants may have exaggerated or minimized their social media or work procrastination behavior in order to make a better impression.

### **Cross-Sectional Design**

In this research, we used cross sectional method for data collection that means all data is collected at a single point of time. Therefore, we did not find cause-effect relationship between the variables. Also, we wouldn't be able to find change in variable over the time.

### **Sample Heterogeneity**

Although the sample was broad, including school, college, and university students, these groups are likely to be very different in terms of academic demands, learning environment, and self-regulation capacity that made it difficult for us to find direct relationships between the variables. These relationships have been observed in the original population with a more homogeneous sample.

### **Narrow Construct Measurement**

In this Research, each construct was measured by only five items. Using scales with more items and a wider range of factors that provides us with better understanding of relationship between self-regulated learning and academic procrastination. Using boarder measurements tools in future research might enhance the accuracy and quality of the statistical analysis.

### **Lack of control variables**

In this study, there were no controlling variables that control the academic behavior of students. Factors such as academic motivation, parents' presence, home environment, mental health and workload. These factors might affect social media addiction and academic procrastination. For future research we should consider these factors for effective and efficient analysis.

## 5.5 Recommendations for Future Research

- **Larger and More Homogeneous Samples:**

Future studies should go after more specific student populations — like vocational learners, university students, or even particular academic disciplines . That way the comparisons can be more focused, not just broad, and the statistical power gets better, hopefully.

- **Longitudinal Design:**

If researchers follow the same students across several academic semesters in a longitudinal study , they can better untangle which effects are causal, not just correlational, and then see how the back and forth between social media addiction, self regulation, and procrastination keeps shifting. Over time , it becomes easier to spot the way these elements interact, rather than seeing only a single slice of it.

- **Additional Variables:**

Future research should bring in more relevant constructs like academic motivation , digital self-efficacy , parental support , and smartphone addiction (not really the same as social media addiction) so that the model becomes a more encompassing explanatory framework.

- **Mixed Methods Approach:**

Qualitative data gathering, like interviews or even focus groups, would pair up with the quantitative results a bit better, because it gives richer understanding around how students really experience, day to day, their “social media addiction” and the whole procrastination thing. In other words , the numbers can show a pattern but the subjective angle, the lived perspective , helps explain why it happens.

- **Alternative Mediators and Moderators:**

Since the mediation results are non-significant, future research should look into other possible mediators (for example academic motivation, digital literacy) and also potential moderators ( like institutional support, parental monitoring) , because these might end up better explaining the actual mechanism connecting social media addiction to procrastination.

## 5.6 Conclusion

This study aimed to look at how social media addiction, kind of, affects academic procrastination, while self-regulated learning was treated as some kind of mediating variable. The whole thing was done using a mixed group of 160 students from school, college, and university levels. The data were gathered via a structured online questionnaire, and then analyzed with SEM-PLS using SmartPLS 4.1, (not any other tool).

The results give partial backing to the proposed framework. In particular, the direct link between social media addiction and academic procrastination, labeled H1, was shown to be positive and statistically significant, which also matches what prior studies have reported. But then the other proposed paths, from social media addiction to self-regulated learning (H2), from self-regulated learning to academic procrastination (H3), and the indirect mediation result (H4) did not reach statistical significance in this sample. So, the mediation route involving self-regulated learning may be more situational, or even context-sensitive, and it might not work the same way across different student backgrounds.

Even with those non-significant mediation paths, the confirmed direct relationship between social media addiction and academic procrastination still feels important and practical. It suggests that reducing excessive social media use should be viewed as part of wider actions to support better academic involvement and productivity among students. For future work, using larger samples that are more uniform, plus longitudinal designs, will likely be necessary to understand more clearly the pathways through which digital habits end up shaping academic results.

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